Drew University - CSCI 260 - Fall 2024
HW6 - C Programming - Basics

HW Info.: Due on 11/20/2024 (W) by 11:59pm, 20 points
Submission: Submit a single C source file *TwosComp.c* on Moodle. In the very first line of the program, put your name in the comment with //.
Note: Please follow Drew University's Standard of Academic Integrity and refer to the course syllabus for late submission policy.

Program Description

(10 points) **Decimal to 2's Complement**: Implement a program that converts a signed decimal number input from keyboard to its corresponding Two's Complement binary form using a char array.

1. At the beginning, the program will ask the user to input a decimal number that can be positive/zero/negative via the prompt "Enter an integer: ". (Assume that the input is indeed an integer.)
2. Then the program will, first, convert that integer to its corresponding unsigned binary form; and then, if the input was negative, apply 2's complement on the unsigned binary.
3. Lastly, print out the obtained 2's complement binary number. **For example**, if the input was "-104", then after the program executes, the console should contain the text "2's Complement: 10011000"; or if the input was "314", then the console should contain "2's Complement: 0100111010".om
4. Your program should contain sufficient comments and should at least build without any errors. In addition, the variable names should be self-explanatory; e.g., *int quotient;* and *int remainder* instead of *int a;* or *int x;*.
5. Submit a single C source files *TwosComp.c* on Moodle. In the very first line of the program, put your name in the comment with //.

Hints

1. To get an integer input from the keyboard, use the "scanf("%d", &varName);" command. Don't forget the ampersand before varName.
2. You want to use a char array of certain length as the container of the '0'/'1' bits. In order to determine that length, you may want to use the "floor()" and "log2()" functions from the "<math.h>" library.
3. You have the / (division) and % (remainder) operators available.
4. When dealing with strings, don't forget that there should always be an additional entry for the terminating char "\0". You may want to manually set it.
5. Don't forget that you will also need to manually include an additional sign bit.
6. When you are iterating/accessing array elements, note that the most-significant bit in binary is accessed through the 0th. index; whereas the least-significant bit is accessed through the (length-1)th. index.
7. When dealing with 2's complement, recall the **shortcut** approach, so you don't really need to implement the binary addition functionality.